

Databases Laboratory 1 Report

Assignments 1 – 3

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Assignment 1 – Student Information System

Entity–Relationship Model (ERM)

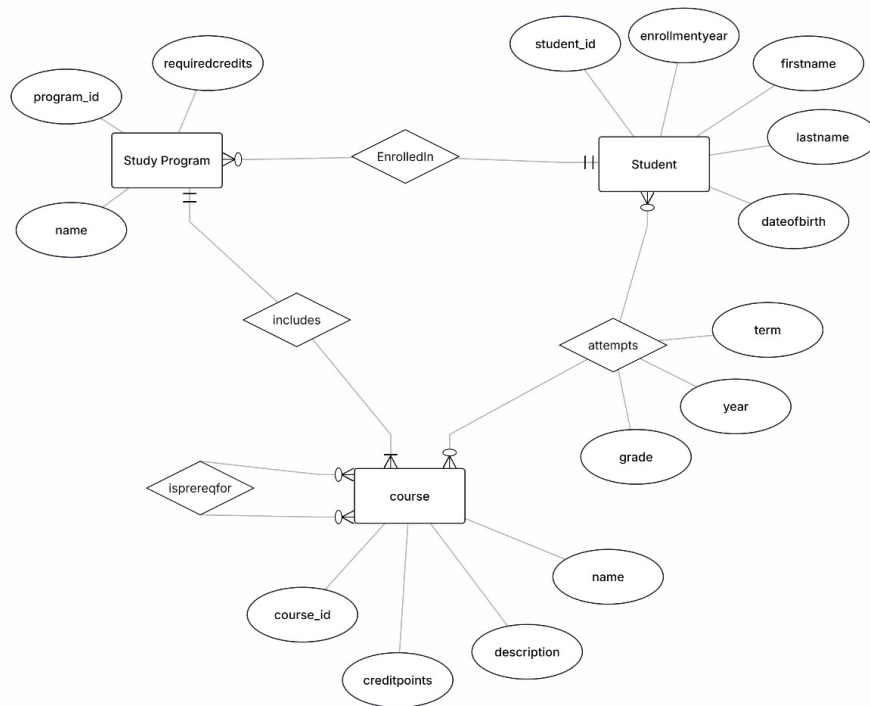


Figure 1: Assignment 1 – ER Diagram

Relational Schema

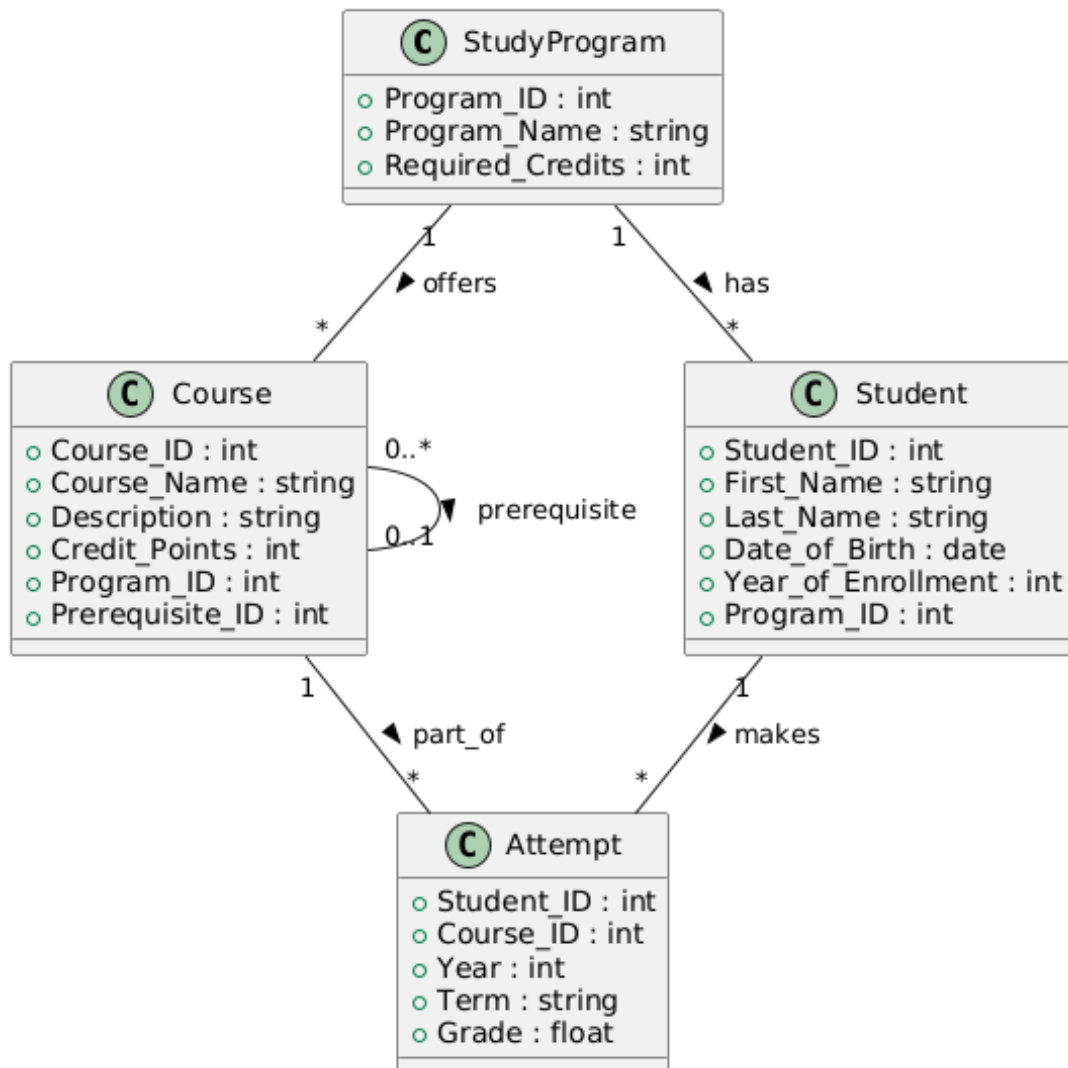


Figure 2: Assignment 1 – Relational Schema

SQL Implementation

```
CREATE TABLE STUDY_PROGRAM (  
    ProgramID VARCHAR(10) PRIMARY KEY,  
    Name VARCHAR(100) NOT NULL UNIQUE,  
    RequiredCredits INT NOT NULL CHECK (RequiredCredits > 0)  
);  
  
CREATE TABLE STUDENT (  
    StudentID SERIAL PRIMARY KEY,    -- SERIAL is an auto-  
        incrementing integer  
    FirstName VARCHAR(100) NOT NULL,  
    LastName VARCHAR(100) NOT NULL,  
    DateOfBirth DATE NOT NULL,  
    EnrollmentYear INT NOT NULL,  
  
    -- Foreign key to link student to a program  
    -- This is NOT NULL because participation is Total  
    ProgramID_FK VARCHAR(10) NOT NULL,  
    FOREIGN KEY (ProgramID_FK) REFERENCES STUDY_PROGRAM(ProgramID  
        )  
);  
  
-- 3. Create COURSE, which references STUDY_PROGRAM  
CREATE TABLE COURSE (  
    CourseID VARCHAR(10) PRIMARY KEY,  
    Name VARCHAR(100) NOT NULL,  
    Description TEXT,  
    CreditPoints INT NOT NULL CHECK (CreditPoints > 0),  
  
    -- Foreign key to link course to a program  
    -- This is NOT NULL because participation is Total  
    ProgramID_FK VARCHAR(10) NOT NULL,  
    FOREIGN KEY (ProgramID_FK) REFERENCES STUDY_PROGRAM(ProgramID  
        )  
);  
  
-- 4. Create the PREREQUISITE junction table  
-- This links two courses together  
CREATE TABLE PREREQUISITE (  
    CourseID_FK VARCHAR(10) NOT NULL,  
    PrereqCourseID_FK VARCHAR(10) NOT NULL,  
  
    -- A course that IS the prerequisite  
    FOREIGN KEY (PrereqCourseID_FK) REFERENCES COURSE(CourseID),  
  
    -- The course that HAS the prerequisite  
    FOREIGN KEY (CourseID_FK) REFERENCES COURSE(CourseID),  
  
    -- The primary key is the combination of both  
    PRIMARY KEY (CourseID_FK, PrereqCourseID_FK)
```

```

);

-- 5. Create the ATTEMPT junction table
-- This links students to courses they've taken
CREATE TABLE ATTEMPT (
    StudentID_FK INT NOT NULL,
    CourseID_FK VARCHAR(10) NOT NULL,
    Year INT NOT NULL,
    Term VARCHAR(10) NOT NULL CHECK (Term IN ('Summer', 'Winter')),
    Grade INT CHECK (Grade >= 0 AND Grade <= 15),

    -- Foreign keys
    FOREIGN KEY (StudentID_FK) REFERENCES STUDENT(StudentID),
    FOREIGN KEY (CourseID_FK) REFERENCES COURSE(CourseID),

    -- Composite primary key
    PRIMARY KEY (StudentID_FK, CourseID_FK, Year, Term)
);

-- 1. insert study programs
INSERT INTO STUDY_PROGRAM (ProgramID, Name, RequiredCredits)
VALUES
('CS-BSC', 'B.Sc. Computer Science', 180),
('BI-BSC', 'B.Sc. Business Informatics', 180);

-- 2. insert students
-- (The StudentID will be auto-generated: 1, 2, 3)
INSERT INTO STUDENT (FirstName, LastName, DateOfBirth,
    EnrollmentYear, ProgramID_FK)
VALUES
('Jason', 'Kennedy', '2002-05-15', 2023, 'CS-BSC'),
('Emir', 'Doe', '2001-11-30', 2022, 'BI-BSC'),
('Efe', 'Adkins', '2003-01-20', 2023, 'CS-BSC');

-- 3. insert courses
INSERT INTO COURSE (CourseID, Name, Description, CreditPoints,
    ProgramID_FK)
VALUES
('CS101', 'Intro to Programming', 'Learn the basics of Python.',
    6, 'CS-BSC'),
('CS220', 'Data Structures', 'Learn about lists, trees, and
    graphs.', 9, 'CS-BSC'),
('BUS101', 'Intro to Business', 'Basics of business
    administration.', 6, 'BI-BSC'),
('BI300', 'ERP Systems', 'Study of enterprise resource planning.'
    , 9, 'BI-BSC');

-- 4. insert prerequisites
INSERT INTO PREREQUISITE (CourseID_FK, PrereqCourseID_FK)

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VALUES
('CS220', 'CS101');

-- 5. insert attempts
INSERT INTO ATTEMPT (StudentID_FK, CourseID_FK, Year, Term, Grade
)
VALUES
(1, 'CS101', 2023, 'Winter', 13),
(3, 'CS101', 2023, 'Winter', 15),
(2, 'BUS101', 2022, 'Winter', 10),
(1, 'CS220', 2024, 'Summer', 11);

select * from study_program;
select * from student;
select * from course;
select * from prerequisite;
select * from attempt;

TRUNCATE TABLE ATTEMPT, PREREQUISITE, STUDENT, COURSE,
STUDY_PROGRAM
RESTART IDENTITY;

```

Assignment 2 – Technology Support System

Entity–Relationship Model (ERM)

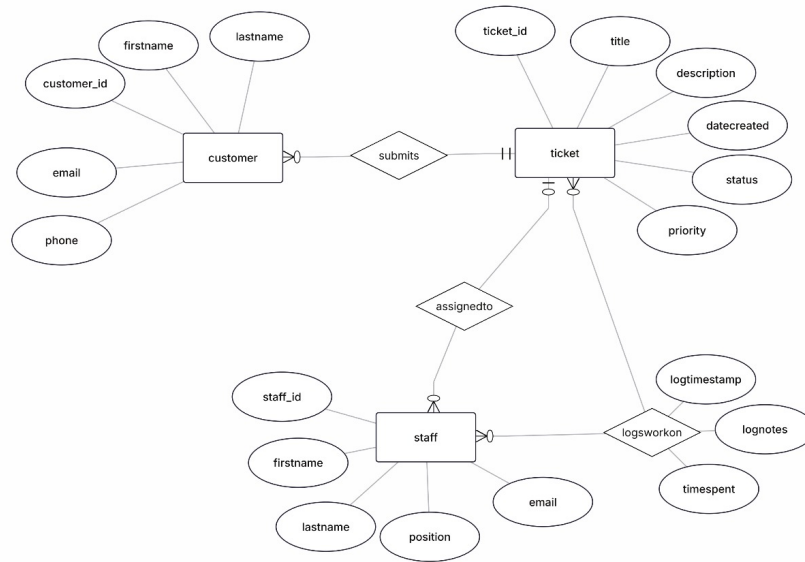


Figure 3: Assignment 2 – ER Diagram

Relational Schema

Technology Support System - ER Diagram

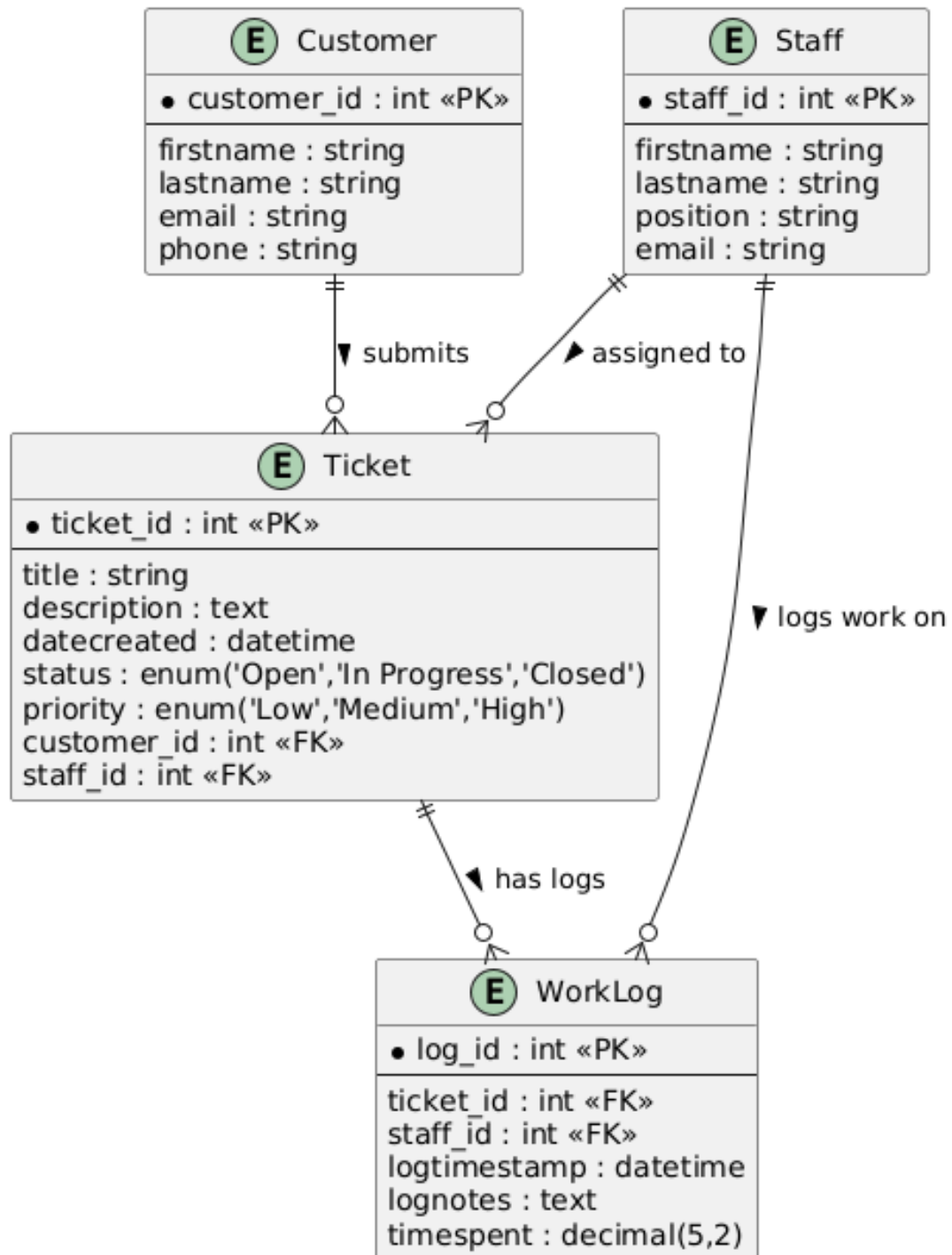


Figure 4: Assignment 2 – Relational Schema

SQL Implementation

```
-- =====
-- INSERT SAMPLE DATA INTO CUSTOMER, STAFF, TICKET, WORK_LOG
-- =====

-- 1      CUSTOMERS
INSERT INTO CUSTOMER (FirstName, LastName, Email, Phone) VALUES
('Anna', 'M ller', 'anna.mueller@example.com', '+49-170-1112233'
),
('Tom', 'Becker', 'tom.becker@example.com', '+49-160-2223344'),
('Lisa', 'Schmidt', 'lisa.schmidt@example.com', '+49-171-3334455'
),
('Felix', 'Hoffmann', 'felix.hoffmann@example.com', '
+49-172-4445566');

-- 2      STAFF
INSERT INTO STAFF (FirstName, LastName, Email, Position) VALUES
('Markus', 'Weber', 'markus.weber@techco.de', 'Technician'),
('Sarah', 'Klein', 'sarah.klein@techco.de', 'Support Engineer'),
('Jonas', 'Bauer', 'jonas.bauer@techco.de', 'Manager');

-- 3      TICKETS
-- Some tickets are assigned, others unassigned
INSERT INTO TICKET (Title, Description, Status, Priority,
CustomerID_FK, StaffID_FK) VALUES
('Printer not working',
'Office printer HP 4020 is offline since morning.',
'Open', 'High', 1, 1),

('Email access issue',
'Customer cannot connect to Exchange server.',
'In Progress', 'Medium', 2, 2),

('Slow internet connection',
'Internet speed drops below 10 Mbps after 2 PM.',
'Unassigned', 'Low', 3, NULL),

('Software installation request',
'Install Microsoft Office 365 on new laptop.',
'Closed', 'Medium', 4, 1),

('Monitor flickering',
'Customer reports screen flickers intermittently.',
'In Progress', 'High', 1, 3);

-- 4      WORK_LOGS
-- These represent the "logs work on" relationship
INSERT INTO WORK_LOG (LogNotes, TimeSpent, StaffID_FK,
TicketID_FK) VALUES
```



```

('Checked printer network connection and restarted print spooler.
', 30, 1, 1),
('Reinstalled HP driver and verified print test page.', 45, 1, 1)
,
('Reset customer mailbox and verified login.', 20, 2, 2),
('Contacted ISP and confirmed line issue; monitoring performance.
', 60, 3, 3),
('Installed Office 365 and configured Outlook account.', 40, 1,
4),
('Replaced HDMI cable and updated GPU driver.', 25, 3, 5),
('Created follow-up ticket for replacement monitor.', 10, 3, 5);

-- 5          CHECK RESULTS
SELECT * FROM CUSTOMER;
SELECT * FROM STAFF;
SELECT * FROM TICKET;
SELECT * FROM WORK_LOG;

```